Joshua Yoo

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WORK EXPERIENCE

Boston Dynamics — **Boston, MA**
Systems Test Engineering Co-op — *January 2026 – Present*
- Develop and run full test cycles for humanoid robot subsystem test stands to validate system performance and reliability
- Capture and analyze high-fidelity subsystem data to drive insights, directly informing design decisions and component-level improvements in humanoid robotics development
- Implement Python-based automated test scripts with closed-loop control and integrated safety mechanisms to improve testing reliability, and data accuracy
- Diagnose and resolve mechanical, electrical, software, and network issues across subsystem test stands to maintain operational uptime and support consistent data collection

Sea Machines Robotics — **Boston, MA**
Software Engineering Intern — *January 2025 – July 2025*
- Applied comprehensive knowledge of hardware, software, and network compatibility to design and implement control systems for Unmanned Surface Vessels (USV)
- Configured and calibrated computer vision systems, optimizing network, and camera settings to deliver low-latency, high-quality imaging for customer-ready plug-and-play deployment
- Designed and implemented a standardized wide-angle camera calibration process that minimized distortion error, enhancing object detection accuracy in USV computer vision systems by 45%

Siemens — **Boston, MA**
Systems Specialist — *July 2022 – September 2024*
- Installed, troubleshot, and commissioned building automation for diverse HVAC systems in multi-million dollar complexes like pharmaceuticals, hospitals, high-rise buildings, and campuses
- Integrated TCP/IP controllers into mission critical networks while maintaining security standards and operation reliability
- Leveraged data-driven insights to identify and implement quality-improvement initiatives that enhanced system efficiency and reduced energy costs by $900K annually across multiple customer-sites
- Served as a customer-facing technical liaison, managing client relationships and communicating complex system requirements and deliverables to both technical and non-technical stakeholders

EDUCATION

Northeastern University — **Boston, MA**
Master of Science in Mechanical Engineering, Mechatronics — *September 2024 – May 2026*
- *Relevant Coursework:* Control Systems Engineering, Robot Mechanics and Control, Mechatronics Systems, Mobile Robotics

Gordon Institute of Engineering Leadership Certification — *September 2024 – August 2025*
- *Concepts Covered:* Technical Leadership, Six Sigma, Product Development, Systems Design, Project Management

Massachusetts Maritime Academy — **Cape Cod, MA**
Bachelor of Science in Energy Systems Engineering — *September 2018 – June 2022*

PORTFOLIO PROJECTS

Autonomous Warehouse Robotics Platform, Personal Project — *September 2025 – December 2025*
- Design and build a scaled-down warehouse logistics system with multiple Automated Guided Vehicles (AGV) and a robotic arm for payload handoff and transfer
- Implement a Mosquitto MQTT broker to manage distributed communication across mobile robots, transmitting optimized waypoint routes and maintaining shared positional awareness via grid-mapped localization using April tag detections

Optimized and Standardized Camera Calibration for USV Navigation, Sea Machines Robotics — *January 2025 – July 2025*
- Optimized the camera calibration pipeline to consistently achieve sub-pixel reprojection accuracy, enhancing vision-based navigation
- Implemented optical and computational methods including Laplacian variance–based focus quantification, and iterative refinement of calibration parameters to ensure uniform precision across the full field of view
- Engineered a networked camera calibration architecture with camera data streamed via RTSP to a processor running ROS2 as middleware, integrated a Python-based back-end server for processing detections, and developed a browser-accessible HTML interface for live visualization, tag mapping, and calibration monitoring

TECHNICAL SKILLS

Programming: Python, C++, MATLAB
Hardware: GD&T, 3D Printing, Arduino, Raspberry Pi, Electrical Troubleshooting
Software: CATIA, SOLIDWORKS, AutoCAD, Git, OpenCV, fastAPI
Communication Protocols: EtherCAT, CAN, TCP/IP
Operating Systems: Linux (Ubuntu/Debian-based), Windows, ROS2